

Beyond the Axion

What else can a worldwide fleet of idle NMR spectrometers measure?

A five-lens discovery pass proposed 28 candidate physics targets; every one was adversarially re-derived from the hardware facts and graded against the strongest published bound. This is the vetted result — including what died, and why.

Verdict	Count	Meaning
Flagship	1	adopt into the core observing program
Strong byproduct	8	real physics reach for Tier A/B cost
Cheap byproduct	6	worth the marginal analysis, modest reach
Gated / deferred	6	viable only after a specific gate is passed
Rejected, on record	7	independently re-derived kill, kept so it is never re-proposed

Cost tiers. **A** = pure re-analysis of bits the network already records; **B** = software/protocol change only, still runnable overnight by facility staff; **C** = cheap per-site hardware or samples. Anything needing magnet or probe modification was rejected outright.

1 Adopt into the core program

1.1 Lock-step spectral tiling

FLAGSHIP TIER A COMPETITIVE NICHE

Stop sitting at one mass point per magnet. Step the ^2H lock offset through ~ 27 positions a night and stitch in absolute-frequency space: the $\pm\text{few-kHz}$ lock shift multiplies by $\gamma_{\text{H}}/\gamma_{\text{D}} = 6.51$ at the ^1H channel, so each site tiles **$\pm 13\text{--}26\text{ kHz}$ of contiguous mass coverage** instead of an isolated 300 Hz sliver. The step pattern is also a veto no fixed-frequency haloscope has: spin lines move with the step, instrument lines do not, and a real dark-matter line stays put in absolute frequency while its amplitude follows the resonant envelope.

27 bins/night costs $27^{1/4} = 2.3\times$ per-bin depth for $27\times$ coverage -- scan figure-of-merit exactly neutral. OCXO $10^{-7}\text{--}10^{-8} = 6\text{--}60\text{ Hz}$ at 600 MHz, already adequate vs the 300 Hz virial width. Lock settle + T1: seconds vs $\sim 20\text{ min}$ segments.

Needs: per-step receiver-carrier follow, per-segment lock-offset metadata in the bundle, stitching in analysis. A direct upgrade of the shipped v0.5 sweep feature.

1.2 Cross-site mass-point overlap and veto protocol

TIER A COMPETITIVE NICHE

Coincidence bookkeeping across sites suppresses accidental line candidates by $10^2\text{--}10^4$. Review corrected two assumptions: the console OCXO already supports cross-site mass matching to $\pm\text{one}$ virial linewidth (a GPSDO is optional hardening, not a prerequisite), and only site pairs at the bottom

of the same-nominal-field spread (60–600 kHz at 600 MHz, vs ± 13 –26 kHz stepping reach) can be brought onto the same mass point.

Needs: lock-offset metadata, a cross-site persistent-line catalog, absolute-frequency bookkeeping discipline in the pipeline.

2 Free analysis riders — same recorded bits

Four searches that ride the exact bytes the network already uploads. Zero observing time, zero hardware.

2.1 Axial-vector dark matter reinterpretation

TIER A WORLD-LEADING

The single best find of the pass. A light vector with a direct axial spin coupling (a “dark Z' ”) drives the *identical* candidate lines as the axion wind, but without the $v \sim 10^{-3}$ velocity suppression. Every core limit converts by one factor.

$g_A = g_{aNN} m_a v \Rightarrow$ core reach $g_{aNN} \sim 10^{-8} \text{ GeV}^{-1}$ gives $\mathbf{g_A} \sim 2.5 \times 10^{-26}$ at $2.5 \mu\text{eV}$. No direct laboratory limit exists anywhere in the μeV decade; this sits 4–8 orders below even the indirect meson-decay bounds.

Needs: one coupling-conversion factor and a modified daily-modulation template in the analysis code. Frame as a limit — UV completions of light non-conserved axial vectors are contrived, so never as a discovery favorite.

2.2 In-bore dark-photon line search

TIER A COMPETITIVE NICHE

Dark-photon dark matter drives an effective current inside the detection volume; the below-cutoff bore blocks external RFI but not this in-volume source, so every probe is a small shielded haloscope. Review corrected the first-pass power estimate by $\sim 300\times$ (a sub-wavelength shielded pickup pays $(R/\lambda)^2 \approx 0.016$ at 600 MHz) and shrank the “virgin territory” claim after checking historical coverage: original ADMX scanned 460–890 MHz in 2010, CAPP-12TB covered 1.025–1.185 GHz.

$\varepsilon \sim 1.4 \times 10^{-11}$ per cryoprobe-night, $\sim 3 \times 10^{-12}$ **per cryo site-year**: wins 3 – $30\times$ over WISPDMMX ($\sim 10^{-11}$) and cosmology ($\sim 10^{-10}$) in the genuinely unscanned windows — ~ 400 – 460 MHz (the 9.4 T sites) plus gaps between historical scans.

Needs: console spurs are *also* immobile under lock stepping (the lock moves B_0 , not the synthesizer) — spur discrimination rests on the clock-audit-keyed spur catalog, blank-tube nights, and optionally a GPSDO (tightens the absolute-frequency test from ~ 60 Hz to < 1 Hz).

2.3 High-resolution sub-virial channel

TIER A COMPETITIVE NICHE

A second FFT pass at ~ 0.1 Hz resolution buys 2.5 – $7\times$ in coupling on cold-flow and caustic substructure — ADMX runs the same high-resolution trick for the photon coupling; nobody does it for the gradient coupling at these mass points. Non-negotiable detail from review: Earth’s rotation against a ~ 300 km/s

flow chirps the line by ~ 0.9 Hz over a night, more than its width, so *diurnal* chirp templates are mandatory, not just annual ones.

Needs: second FFT pass, diurnal+annual template bank, lock-step amplitude veto against the sub-Hz instrumental spur forest. OCXO stability floors usable linewidth at ~ 0.06 Hz.

2.4 Calibrated excess-power archive (public data product)

TIER A FIRST-LIMIT TERRITORY

A versioned public release: calibrated PSDs + covariance + spur list + measured coil/shield transfer function (including the sub-Compton shielding suppression) + field, orientation, coordinates per site. On every model anyone has written down we lose 1–3 orders to dedicated experiments — but the archive makes the dark-photon, axion–photon, and high-frequency gravitational-wave reinterpretation tables free forever, and in our exact slivers it is the only limit of any kind.

3 Worth a protocol change (Tier B)

3.1 Sidereal Lorentz violation on the clock-blind $^1\text{H}/^2\text{H}$ ratio

TIER A/B WORLD-LEADING

Both frequencies come from the same OCXO, so oscillator and B_0 cancel exactly in the ratio — and the lock servo is the mechanism, not a bug: it transduces any deuteron-sector shift into B_0 and hence into the ^1H channel. The *B-scaled* channel (an anisotropy of the bound proton/deuteron magnetic moment, $\delta g \propto B$) is virgin territory that high field actively helps.

Statistical: $\sim 2 \times 10^{-13}$ per network-year. Systematics haircut (spin-noise line pull needs a demonstrated 10^2 – 10^3 sidereal-coherent suppression): floor 3×10^{-12} – 10^{-11} fractional — still beats BASE’s proton sidereal sensitivity ($\sim 10^{-9}$ at 1.9 T) by **100–1000**×, with $\times 7$ – 15 extra coefficient reach at 14–28 T. Conceded: the B-independent $\tilde{b}$ channel loses 10^4 – 3×10^7 to comagnetometers — dead, see the ledger.

Needs: standardized 90/10 $\text{H}_2\text{O}/\text{D}_2\text{O}$ network sample, per-block lock error-signal logging, phased-lineshape co-fit, detuning-ladder validation (already planned for core), an SME theory workup for the deuteron.

3.2 Isotope-ladder X-nuclei campaign — ^{19}F adopted

TIER B FIRST-LIMIT TERRITORY ^{19}F ADOPTED

^2H and ^{13}C nights give the first *laboratory* limits on the axion–neutron coupling at these mass points (^{31}P is odd-proton — a proton-coupling mass point on the X channel, not a neutron probe); ^{19}F at $0.941 \times$ the ^1H frequency rides the $^1\text{H}/^{19}\text{F}$ coil at near-cryo sensitivity — essentially free proton mass points densifying the scan. Review killed a wrong $\gamma^{1.5}$ scaling (the X channel is circuit-noise-limited; real numbers are 10–30× worse than first proposed) and the campaign survives as lab-frontier exclusion, honestly above SN1987A.

$g_{ann} \sim 2 \times 10^{-5}$ – 2×10^{-4} per season per mass point via ^2H . Protocol rules that survived: ^2H observation **requires lock-off nights** (the lock transmitter irradiates the deuterium line; drift 0.1 Hz/hr is negligible vs the 46 Hz virial width); ^{13}C needs overnight decoupling or $^{13}\text{CS}_2$ against the 140 Hz J-quartet.

Decision, 2026-09-01: the ^{19}F rider is **adopted** into the near-term program, split off from the rest of the ladder — it needs no X-channel calibration work, a C_6F_6 tube is cheap and inert, and every

F-capable site that runs it densifies the proton mass scan for free. The $^2\text{H}/^{13}\text{C}$ neutron-coupling legs (and the ^{31}P proton-channel leg) stay Tier B pending volunteer sites.

Needs: ATMA retune scripts, per-nucleus calibration, $\sim \$100\text{--}500$ in samples (neat D_2O , ^{13}C -methanol or $^{13}\text{CS}_2$, C_6F_6 , 85% H_3PO_4) — all standard 5 mm tubes.

3.3 Dual-species transient network (GNOME complement)

TIER B COMPETITIVE NICHE

Interleaved $\sim 1^\circ$ FIDs at 0.5–1 Hz on a $\sim 1\%$ H_2O / 99% D_2O sample track the ^1H line at $\sim 1\text{ MHz/s}$ — 23 pT proton-equivalent, the same decade as GNOME’s effective *proton*-coupling floor (alkali magnetometers pay $\sim \mu_N/\mu_B$ against nuclear couplings) — with exact γ_d/γ_p dual-species tagging for domain-wall candidates. The systematic that sets the dilution: at 10% H_2O , radiation damping pulls the line at $1000\times$ the target; at 1% it is controllable, pending demonstration.

Needs: $\geq 10\text{ Hz}$ lock-error logging, the interleaved-FID pulse program, lock transfer-function calibration via existing lock-shift stepping, a GNOME-style coincidence pipeline on NTP timestamps.

3.4 Foundational pair: FDT at GHz and collective-decay metrology

TIER A/B COMPETITIVE NICHE

(a) First quantitative fluctuation–dissipation test on a nuclear-spin ensemble at GHz: calibrated spin-noise PSD vs small-flip χ'' at the $\sim 1\%$ absolute-calibration floor, including the two-temperature cryoprobe case (20 K coil, 300 K spins). Verified unoccupied territory. The quantum-asymmetry stretch goal is dead: a phase-insensitive amplifier measures the symmetrized spectrum, correction $(hf/2kT)^2/3 \sim 2 \times 10^{-7}$. (b) Absolute radiation-damping metrology: the $\omega Q \eta M_0$ collective-decay rate tested at few-percent absolute across $2\times$ in frequency and $100\times$ in M_0 on many platforms — $\sim 10\times$ beyond the single-instrument literature. Frame as coupled-mode physics, not “Purcell,” or referees will object.

4 Gated or deferred

4.1 Axion–photon piggyback — gated on one number

GATED TIER A + 1 WK SIM

The 9.4–28 T field through the detection volume makes every probe a (terrible) haloscope — but an NMR coil is engineered to have zero leading-order overlap with the axisymmetric conversion field, and the residual form factor C (lead routing, tilts, dielectric asymmetries) is unknown between 10^{-2} and 10^{-5} . The gate is ~ 1 week of full-wave EM simulation of the 2–3 dominant commercial probe geometries.

$C \geq 10^{-2}$: $g \sim 2 \times 10^{-11} \text{ GeV}^{-1}$ per cryo-site-year — **beats CAST/globular clusters** ($6.6 \times 10^{-11}/4.7 \times 10^{-11}$) **by** $\sim 3\times$ in the $1.65\text{--}2.7 \mu\text{eV}$ window no cavity experiment has ever scanned. $C \sim 10^{-3}$: a tie — publish the simulation and stop. Claim nothing until C is in hand.

4.2 Idle-MRI fleet nodes (64–298 MHz)

PILOT FIRST TIER B, NEW INSTRUMENTS

The 10 L volume win survives honest accounting of coil dilution and room temperature: $\sim 5\times$ one cryoprobe-night per scanner-night in amplitude SNR, at $0.26\text{--}0.53\text{ }\mu\text{eV}$ mass points nothing else covers. A fleet-year reaches $\text{few} \times 10^{-8}\text{--}10^{-7}\text{ GeV}^{-1}$ — the closest approach to the SN1987A bound anywhere in the program. Real costs are institutional: raw-FID console access, a $\sim 10\text{ L}$ phantom, per-scanner characterization. Pilot on 2–3 research scanners before any fleet claim.

4.3 Also parked

- **CASPEr-electric-lite** (poled ferroelectric ^{207}Pb stick, $\sim \$100\text{--}500$): arithmetic survives, but disorder broadening in relaxors costs another $3\text{--}10\times$; pilot-grade, non-cryo probe first.
- **Broadband X-channel scan**: the $(R/\lambda)^2$ suppression is fatal below $\sim 300\text{ MHz}$ (loses $100\text{--}1000\times$ to Dark E-field Radio); only the $300\text{--}460\text{ MHz}$ corner on $\geq 23.5\text{ T}$ systems survives, and $\sim 0.5\text{ TB/night}$ storage dominates the cost.
- **GPSDO clock-defect search**: scientifically near-empty (at transient timescales a GPSDO output is its own local quartz) — but buy the $\sim \$300\text{--}1000$ GPSDO anyway for spur discrimination and cross-night stitching.
- **Driven-lock nonequilibrium FDT**: real physics, but the ^2H noise feature is $\sim 10^{-2}$ of the ^1H bump; needs raw lock-receiver access. Park it.

5 Considered and killed — the ledger

Every rejection was independently re-derived before being recorded. The ledger exists so these are never re-proposed.

- **ULF dark photon via the lock / persistent-current channel** lose by >13 orders to SuperMAG
A sub-wavelength pickup pays (L/λ) in field: 6×10^{-12} at mHz. SuperMAG’s Earth-radius loop has a 2×10^7 head start no lab apparatus buys back.
- **Monopole–dipole force with an in-bore source mass** lose by 7–8 orders to $^3\text{He}/^{129}\text{Xe}$
 0.5 kg at 10 cm reaches $g_s g_p \sim 10^{-22}$ vs comagnetometer bounds at $10^{-29}\text{--}10^{-30}$. Geometry cannot rescue it.
- **B-independent sidereal $\tilde{b}$ on the $^1\text{H}/^2\text{H}$ ratio** lose by $10^4\text{--}3 \times 10^7$
Systematic floor $2.5 \times 10^{-26}\text{ GeV}$ vs bounds down to 8.4×10^{-34} . The deuteron-combination escape fails: $\mu_d \approx \mu_p + \mu_n$, triangle inequality closes it. Only the B-scaled channel lives.
- **Hz–kHz stochastic ALP sidebands / distant-site cross-correlation** lose by 2–6 orders
Best case $g_p \sim 10^{-4}\text{ GeV}^{-1}$ vs NASDUCK-class. And the canonical answer to “why not cross-correlate distant sites?”: the DM coherence length at μeV masses is $\sim 500\text{ m}$.
- **Astrophysical burst follow-up (SN axions, FRBs, solar)** in-band weight $\leq 10^{-27}$
 MeV supernova axions oscillate at 10^{22} Hz ; in-band FRB EM is attenuated e^{-35} by the below-cutoff bore and loses to thermal noise even unshielded.
- **Pauli-exclusion violation in NMR** lose by $\geq 10^9$
Wrong-symmetry admixtures in liquids produce no new frequencies — only an intensity shift degenerate with concentration. Best conceivable bound $\sim 10^{-2}$ vs spectroscopy at 10^{-11} .

- **Millicharged-particle cyclotron lines** conditions mutually exclusive
In-band cyclotron and terrestrial capture cannot both hold: they miss each other by >an order of magnitude in ε at every mass.
- **Oscillating quark masses (dilaton DM) in the ratio** MICROSCOPE wins by ≥ 3.5 orders
Best-case $d_{m_q} \sim 1.6$ at the single most favorable mass. The ratio time series survives as a free drift diagnostic only.
- **High-frequency gravitational waves as a physics driver** $h \sim 6 \times 10^{-18}$, sources $< 10^{-27}$
Survives only as a free reinterpretation table in the excess-power archive.

Three rules the kills taught us

- **Sub-wavelength EM pickups pay (L/λ) in field** — an $O(0.1)$ penalty at 600 MHz, a death sentence at mHz. Low-frequency ambitions belong to spin-coupled channels.
- **The μeV dark-matter field is coherent over $\sim 500\text{ m}$** — same-hall pairs are phase-coherent; the worldwide network is not. Its power is mass points and vetoes, not coherent combining.
- **Phase-insensitive amplifiers measure the symmetrized spectrum** — linear-in- hf/kT quantum asymmetries are invisible; the accessible correction is $(hf/2kT)^2/3$.

6 Near-term actions

In rough order of leverage. Items 1–4 are pure software.

1. **Lock-step tiling mode** — per-step carrier follow, per-segment lock-offset metadata, stitching support. [Orchestrator + analysis](#); v0.6 candidate.
2. **Per-block ^2H lock error-signal and lock-reference logging** — wanted independently by four candidates and the core clock audit. [Orchestrator + bundle schema](#).
3. **Analysis riders** — dark-photon line search, axial-vector limit curve, sub-virial pass with diurnal chirp templates, spur catalog keyed to the clock audit. [Analysis repo](#).
4. **Excess-power archive format** — specify the versioned public release. [Analysis + docs](#).
5. **^{19}F pilot (adopted 2026-09-01)** — survey F-capable probes in the site registry, ship C_6F_6 tubes, run pilot noise nights at 2–3 sites, then fold ^{19}F into the standard rotation. [Coordination + volunteer sites](#).
6. **Probe form-factor EM simulation** — the one week of work that decides whether “axion–photon” goes in a proposal. [Any partner with EM-simulation tooling](#).
7. **Standardized network samples** — 90/10 and 1%/99% $\text{H}_2\text{O}/\text{D}_2\text{O}$; remaining isotope-ladder kit for volunteer sites. [Coordination](#).
8. **Optional per-site GPSDO** ($\sim \$300\text{--}1000$) — justified by spur discrimination and cross-night stitching, not by the core search. [Volunteer sites](#).

Produced 2026-09-01 with AI assistance: five independent generation lenses, 28 candidates, each adversarially re-derived against published bounds by a separate review pass. All numbers are order-of-magnitude estimates for prioritization — re-derive independently before any external use. The authoritative source of this document is `docs/science_roadmap.md` in the repository; the full 28-entry catalog with complete assessments is retained by the maintainer.